# Hundred Days of Code
# day 28

QUESTION: ) Write a C program to: a) Read the number of rows and columns from the user. Ensure that both values are positive. If not, repeat to get the desired input(s). b) Declare a variable-length 2D integer array using the given dimensions. c) Read the array elements column by column using nested loops, with the outer loop iterating over columns and the inner loop over rows. d) Print the elements in row-major order using a single loop and a linear index.

2) Write a C program that reads elements into a variable-length 2D array, copies them into a 1D array in row-major order, and displays both arrays.

SOLUTION:

1)

**1. Aim**

To write a C program to accept positive values for the number of rows and columns, declare a **variable-length 2D integer array**, read its elements **column by column**, and display the elements in **row-major order using a single loop and linear index**.

**2. Algorithm**

1. Start the program.
2. Input the number of rows.
3. If the number of rows is less than or equal to zero, repeatedly ask the user to enter a positive value.
4. Input the number of columns.
5. If the number of columns is less than or equal to zero, repeatedly ask the user to enter a positive value.
6. Declare a variable-length 2D integer array arr[rows][cols].
7. Use nested loops to input the elements:
    - Outer loop iterates over columns.
    - Inner loop iterates over rows.
8. To display the elements in row-major order, use a single loop from 0 to rows * cols - 1.
9. For each linear index i, access the corresponding element using:
    - Row index = i / cols
    - Column index = i % cols

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10. Display the elements.

11. Stop.

C Program:

```
#include <stdio.h>


int main()
{
   int rows, cols, i, j;


   // Input and validate number of rows
   do
   {
      printf("Enter number of rows: ");
      scanf("%d", &rows);


      if(rows <= 0)
         printf("Rows must be positive. Try again.\n");


   } while(rows <= 0);


   // Input and validate number of columns
   do
   {
      printf("Enter number of columns: ");
      scanf("%d", &cols);


      if(cols <= 0)
```

```
        printf("Columns must be positive. Try again.\n");

    } while(cols <= 0);

    // Variable-Length 2D Array
    int arr[rows][cols];

    // Input elements column by column
    printf("\nEnter array elements column by column:\n");

    for(j = 0; j < cols; j++)
    {
       for(i = 0; i < rows; i++)
       {
          printf("Element [%d][%d]: ", i, j);
          scanf("%d", &arr[i][j]);
       }
    }

    // Display in row-major order using a single loop
    printf("\nElements in row-major order:\n");

    for(i = 0; i < rows * cols; i++)
    {
       printf("%d ", arr[i / cols][i % cols]);

       if((i + 1) % cols == 0)
```

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```
        printf("\n");
    }


    return 0;
}
```

Test Case Table :

| Test Case | Rows | Columns | Input Elements | Expected Result |
|---|---|---|---|---|
| 1 | 2 | 3 | 1, 2, 3, 4, 5, 6 | Accepts input and displays a 2×3 array |
| 2 | 3 | 2 | 10, 20, 30, 40, 50, 60 | Accepts input and displays a 3×2 array |
| 3 | 1 | 1 | 5 | Displays 5 |
| 4 | 0 | 3 | — | Asks user to re-enter rows |
| 5 | 2 | -1 | — | Asks user to re-enter columns |

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Compilation and Execution:



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2)

**1. Aim**

To write a C program to read elements into a **variable-length 2D array**, copy all its elements into a **1D array in row-major order**, and display both arrays.

**2. Algorithm**

1. Start the program.
2. Input the number of rows and columns.
3. Check whether both dimensions are positive.
4. If either value is invalid, ask the user to enter it again.
5. Declare a variable-length 2D array arr[rows][cols].
6. Declare a 1D array of size rows * cols.
7. Use nested loops to input the elements of the 2D array.
8. Traverse the 2D array row by row.
9. Copy each element into the 1D array using a separate linear index k.
10. Display the original 2D array.
11. Display the resulting 1D array.
12. Stop.

C Program:

```
#include <stdio.h>


int main()
{
   int rows, cols, i, j, k = 0;


   // Input and validate rows
   do
   {
      printf("Enter number of rows: ");
```

```
    scanf("%d", &rows);

    if(rows <= 0)
        printf("Rows must be positive. Try again.\n");

} while(rows <= 0);

// Input and validate columns
do
{
    printf("Enter number of columns: ");
    scanf("%d", &cols);

    if(cols <= 0)
        printf("Columns must be positive. Try again.\n");

} while(cols <= 0);

// Variable-Length Arrays
int arr[rows][cols];
int oneD[rows * cols];

// Input elements into 2D array
printf("\nEnter elements of the 2D array:\n");

for(i = 0; i < rows; i++)
{
```

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```
    for(j = 0; j < cols; j++)
    {
        printf("Element [%d][%d]: ", i, j);
        scanf("%d", &arr[i][j]);
    }
}

// Copy 2D array into 1D array in row-major order
for(i = 0; i < rows; i++)
{
    for(j = 0; j < cols; j++)
    {
        oneD[k] = arr[i][j];
        k++;
    }
}

// Display 2D array
printf("\n2D Array:\n");

for(i = 0; i < rows; i++)
{
    for(j = 0; j < cols; j++)
    {
        printf("%d ", arr[i][j]);
    }
    printf("\n");
```

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```
    }

    // Display 1D array
    printf("\n1D Array in row-major order:\n");

    for(i = 0; i < rows * cols; i++)
    {
        printf("%d ", oneD[i]);
    }

    printf("\n");

    return 0;
}
```

Test Case Table:

| Test Case | Rows | Columns | 2D Array Elements | Expected 1D Array |
|---|---|---|---|---|
| 1 | 2 | 3 | 1 2 3 / 4 5 6 | 1 2 3 4 5 6 |
| 2 | 3 | 2 | 10 20 / 30 40 / 50 60 | 10 20 30 40 50 60 |
| 3 | 1 | 4 | 2 4 6 8 | 2 4 6 8 |
| 4 | 1 | 1 | 25 | 25 |
| 5 | -2 | 3 | — | Asks user to re-enter rows |

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Compilation and execution:

